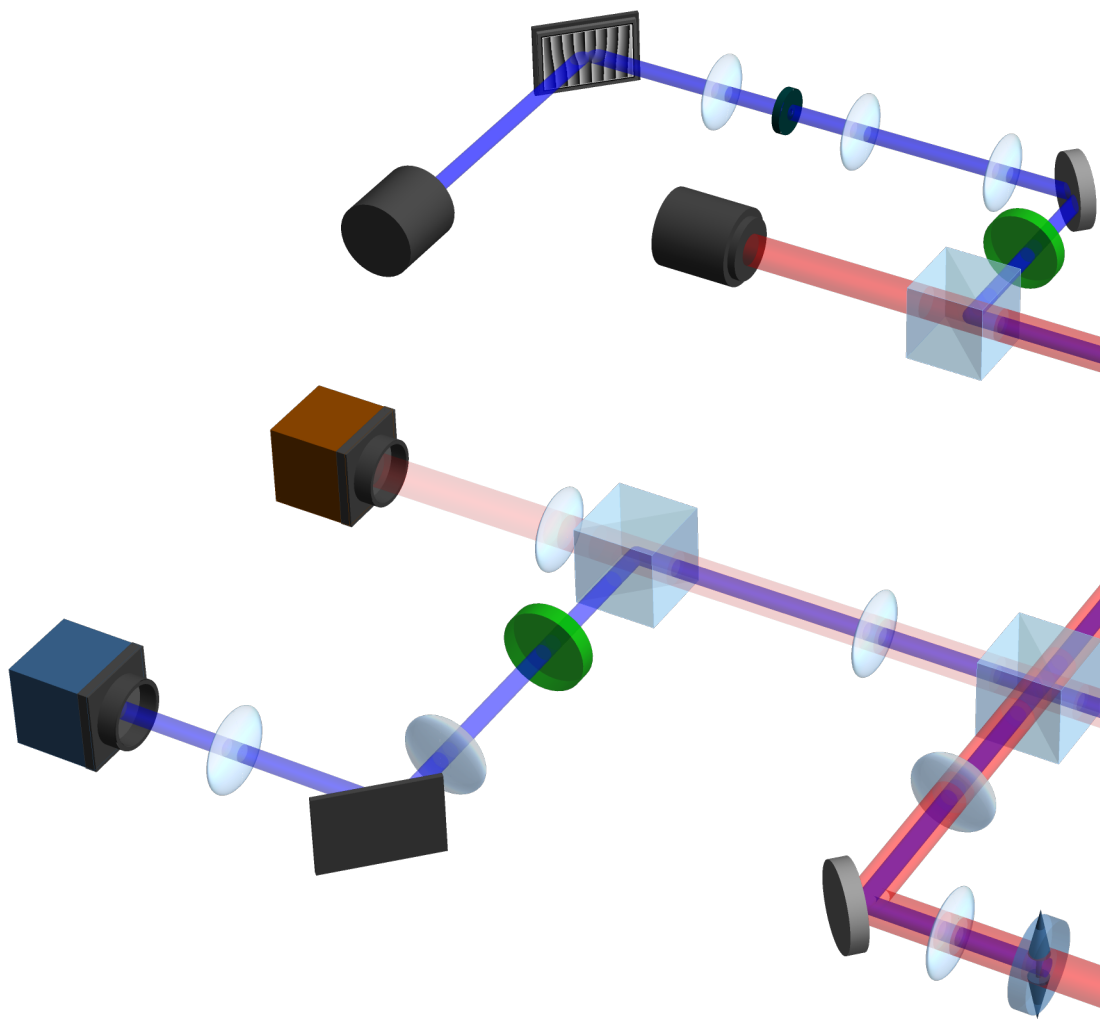
In[3]:=

```
(*Establishes the directory where the notebook is located as the default one*)
SetDirectory@NotebookDirectory[];
<< "OpticalElementDictionary-March2026.m"
```

```
RefBeam = Graphics3D[{{CapForm["Square"], Opacity[0.3], Red,
     Tube[{{1, 0, 0.5}, {8, 0, 0.5}, {8, -10, 0.5}, {12, -10, 0.5}}, 0.2]}, {CapForm[
      "Square"], Opacity[0.1], Red, Tube[{{-1, -6, 0.5}, {11, -6, 0.5}}, 0.2]}}];
OAMBeam = Graphics3D[
   {{CapForm["Square"], Opacity[0.4], Blue, Tube[{{-3, -1, 0.5}, {-3, 3, 0.5}, {4.1,
        3, 0.5}, {4.1, 0, 0.5}, {8, 0, 0.5}, {8, -10, 0.5}, {10, -10, 0.5}}, 0.1]},
    {CapForm["Square"], Opacity[0.3], Blue,
     Tube[{{11, -6, 0.5}, {3, -6, 0.5}, {3, -11, 0.5}, {-0.5, -11, 0.5}}, 0.1]}}];
Show[
 LaserCyl[{0, 0, 0}, 0], (*Red Laser*)
 M[{8, 0, 0}, π / 4],     (*Mirror to the Beam splitter*)
 BS[{8, -6, 0}, 0],      (*Beam Splitter in the AO*)
 Lens[{6, -6, 0}],      (*Telescope of the AO*)
 Lens[{2, -6, 0}],
 WFS[{-1, -6, 0}],      (* ALPAO *)
 WFS[{12, -10, 0}, π],   (* ThorLabs*)
 M[{7.8, -10.2, 0}, π / 4],  (*Mirror for the ThorLabs WFS*)
 DefM[{11, -6, 0}, π], (*Deformable mirror*)
 Lens[{9, -10, 0}, 0],
 Pol[{10, -10, 0}, π / 2, 0],   (*Polarizer*)

 (*OAM Elements*)
 LaserCyl[{-3, -2, 0}, π / 2],  (*Blue Source*)
 SLM[{-3, 3, 0}, "hol1.png", 7 π / 4], (*SLM Place holder*)
 Lens[{-1, 3, 0}, 0],          (*4f system for the SLM*)
 Iris[{0, 3, 0}, π / 2],
 Lens[{1, 3, 0}, 0],
 Lens[{3, 3, 0}, 0],
 (*Lens that couples to the mafnification system of the red source*)
 QW[{4, 1.5, 0}, π / 2],         (*  *)
 BS[{4, 0, 0}, 0],
 M[{3.9, 3, 0}, π / 4],
 BS[{3, -6, 0}, π / 2],
 QW[{3, -8, 0}, π / 2],
 Lens[{8, -8, 0}, π / 2],
 Lens[{3, -10, 0}, π / 2],
 SLM[{3, -11.3, 0}, "hol1.png", 3 π / 4], (*SLM Place holder*)
 Lens[{1, -11, 0}, 0],
```

```
CCD[{-1, -11, 0}],


RefBeam,
OAMBeam,
PlotRange → All, ImageSize → 1000, Lighting → "Neutral"]
```

Out[29]=